



Original Article

Exploring the genomic potential of *Kytococcus schroeteri* for antibacterial metabolites against multi-drug resistant *Mycobacterium tuberculosis*

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ABSTRACT

The versatile bacterium *Kytococcus schroeteri* is known for producing putative secondary metabolites. These include antimicrobials and other therapeutically significant compounds. The emergence of antibiotic-resistant pathogens has necessitated the exploration of possible sources for novel drug candidates. This study focuses on the genome mining of *Kytococcus schroeteri* to explore its secondary metabolites through bio-synthetic gene clusters. It analyzes their drug-like properties through *in-silico* methods and evaluates their potential as antimicrobial agents. Eight biosynthetic gene clusters were identified in two strains (H01 and UMB1298) of this bacterium. Among the unique 49 metabolites from these clusters, 13 metabolites were selected according to the Lipinski rule of five. Physicochemical properties, pharmacokinetic analysis, toxicity profiles, and human target predictions of these metabolites were performed, and they were examined for crucial interactions with *M. tuberculosis*'s RpfB protein, the causative agent of latent tuberculosis. All metabolites were predicted to be non-toxic and did not inhibit any human proteins. Several metabolites, including a subset of brasilanes, exhibited both low acute toxicity and promising interactions with GLU292, the critical residue of the RpfB protein. The interaction affinity of the ligands with RpfB was validated by subjecting one of the complexes to a 100 ns MD simulation. The RMSD, RMSF, and binding energy calculations indicated a stable interaction of the ligand with the receptor protein, which raised the possibility of novel drugs to combat antibiotic resistance. The putative metabolites identified in this study not only exhibit molecular properties but also possess characteristics that support physiological compatibility. It also heightens their potential effectiveness as therapeutic antibacterial alternates. The experimental validation of our computational results may open an avenue to explore the potential of *K. schroeteri* for producing novel compounds to combat the antibiotic resistance in *M. tuberculosis*.

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Introduction

Kytococcus schroeteri is a versatile bacterium found as part of the human skin microbiota but also studied for its potential as an endophyte. Endophytic organisms are known to colonize plants symbiotically, producing metabolites with ecological significance and pharmaceutical potential [1]. In human hosts, *K. schroeteri* is primarily a harmless saprophyte residing on the skin and mucous

membranes. It plays a subtle but important role in maintaining microbial balance on the skin surface and contribute to a stable microbiota environment. However, it can become pathogenic under specific circumstances, particularly in individuals with compromised immune systems, such as cancer patients or those undergoing organ transplantation [2]. The bacterium has been identified as a causative agent in cases of prosthetic valve endocarditis, shunt infections, and even pneumonia in neutropenic patients. These infections highlight its ability to transition from a commensal state to an opportunistic pathogen, especially when host defenses are lowered [1].

Kytococcus schroeteri is also present in *Ephedra foliate* (a medicinal shrub), which is reported to produce secondary metabolites [3]. It is a non-motile, gram-positive, & non-encapsulated bacteria that belongs to the Dermacoccaceae family and the Actinomycetes order [4]. Kytococci are catalase-positive and aerobic bacteria, that, on agar, grow as yellow colonies. The Kytococci genus is recognized to include three species: *K. schroeteri*, *K. aerolatus*, and *K. sedentarius* [5]. Actinomycetes are rich in soil and have received attention for studying their secondary metabolites [6]. They are famous for their ability to produce abundant metabolites with therapeutic potential, especially antibiotics and anticancer drugs. Due to the fact that pathogenic bacteria are growing resistant to multiple antibiotics and causing major health problems, scientists are in search of novel broad-spectrum antibacterial compounds with noteworthy antibacterial properties [7].

The arising resistances to many drugs because of the asymptomatic infections of tuberculosis (TB) pose a potentially fatal challenge, because the available medications such as rifampicin, ethambutol, pyrazinamide, and isoniazid are efficient for drug-sensitive tuberculosis [8] while they are insufficient for its strains that have developed complete resistance against many drugs [9]. RpfB protein that aids in dormancy and is located in the bacteria's cell wall, is considered crucial for the growth of bacteria particularly during the latent period among all the targets studies [10].

Computer-aided prediction and characterization of biosynthetic gene clusters (BGCs) with the potential to produce secondary metabolites has successfully contributed to the rational design of novel therapeutics by analyzing the microbial genome [11,12]. Secondary metabolites of microbes are easy & genuine sources and have been used to develop novel therapeutics for the treatment of a variety of diseases, including fungi, parasites, viruses, and, bacteria. The majority of these metabolites belong to a broad variety of chemical classes and have shown antitumor and antibiotic activities [13].

In this study, genome mining of *Kytococcus schroeteri* was performed to investigate the gene-coding potential of this species for the production of putative secondary metabolites. The drug-like properties of metabolites within the bacteria's coding genome were systematically analyzed to assess their potential. Eight biosynthetic gene clusters were identified in both strains of this bacteria. Among the unique 49 metabolites from these clusters, 13 metabolites were selected according to the Lipinski rule of five. Molecular docking and dynamic simulation were employed for metabolite interactions and stability with the resuscitation-promoting factor B (RpfB) protein. Physicochemical properties, pharmacokinetic analysis, toxicity profiles, and human target predictions of these metabolites were performed, and they were examined for crucial interactions with the RpfB protein of *Mycobacterium tuberculosis*, the causative agent of latent tuberculosis.

Materials and methods

Genome sequences retrieval

The genome sequences of *K. schroeteri* strains H01 (contig, GenBank ID: GCA_006542415.1) and UMB1298 (scaffold, GenBank ID: GCA_002847825.1) were acquired from the NCBI database.

Identification of putative BGCs

In order to find out the BGCs, *K. schroeteri* strains H01 and UMB1298 genomes were submitted to AntiSMASH bacterial version 6.0 to figure out uncharacterized bioactive secondary compounds synthesizing gene clusters and their expected chemical structures. To find BGC areas in genomes, AntiSMASH employs a probabilistic method based on a hidden Markov model (HMM) [14]. The PFAM and the MIBiG database were utilized for additional annotations, domain functions, and the metadata on molecular products & BGCs. ClusterBlast and KnownClusterBlast modules utilized to find comparable clusters in the selected genome [15].

Secondary metabolites library preparation

The structures of predicted secondary metabolites were drawn by utilizing the ChemDraw software [16] and saved in mol format for library preparation.

Drug-like properties and evaluation of putative metabolites

The drug-likeness of gene-coding metabolites according to Lipinski's rule was determined [17]. Lipinski's principle is typically used to assess a chemical compound's potential for use as a medication. When a biologically active lead molecule is given priority during drug research, it is pretentious to take the rule into consideration. The drug-like compound should not have H-bond acceptors > 10 and, H-bond donors > 5, in accordance with Lipinski's criterion. In addition, the molecular weight must be less than 500 Daltons, and the log P partition coefficient of octanol-water must not exceed 5 [18].

Pharmacokinetic analysis

The pharmacokinetic prediction of the secondary metabolites was performed by SwissADME [19]. It is a web tool that is used to compute the physicochemical characteristics as well as predict the pharmacokinetic ADMET properties (absorption, distribution, metabolism, excretion, and toxicity). It also assists in predicting the drug-like properties and compounds medicinal chemistry to promote drug discovery [20].

Toxicity analysis and target prediction probability

The metabolites were further analyzed on the basis of toxicity analysis as (cytotoxicity, hepatotoxicity, LD50 value, and carcinogenicity) by the ProTox-2 server [21]. Target prediction probability analysis was performed using the Swiss Target Prediction server [22]. For performing these analyses, the SMILES codes of the compounds were created and utilized as the input files. The outcomes are displayed as scores that range from 0 to 1, with the value 1 denoting query molecule's most likely target.

Validation of docking protocol

To ensure the accuracy of predictions, molecular docking results must be validated in the drug discovery process. Comparing the docked conformation with a recognized crystal structure is a reliable method of validation [23–25]. This process is essential to avoid potential resource waste and guarantee reliable results when developing new drugs. The conformation of our docked complex of RpfB with benzamidine from the designed structure was compared to the X-ray crystallographic structure of RpfB with its co-crystallized benzamidine (PDB ID: 4EMN). We superimposed the co-crystallized structure with the docked protein-ligand complex using PyMOL. By calculating the RMSD scores between the docked ligands and the

reference, the precision of the alignment was evaluated as reported previously [24], a successful docking strategy has an RMSD value of ≤ 2.0 .

Molecular docking analysis

Molecular docking is an important technique that can anticipate the binding abilities of ligands to protein receptors. It is typically carried out to evaluate the binding affinities of secondary metabolites against the drug target [26]. Docking was performed by the MOE (2022.02) software. The template that has the RpfB complex structure as a drug target was retrieved PDB database (PDB ID: 4EMN). MOE carried out the intricate 3D protonation and energy minimization for the downstream analyses. The molecular binding potential of the ligand and receptor was articulated using the protein-ligand docking approach that was performed in the MOE. With MOE, the generalized Born/volume integral (GB/VI) approach was utilized to determine the ligands' binding affinities. The London dG scoring method was used to determine the ligand-binding free energy [27]. The ligand atoms were allowed to roam about freely in the binding pocket. The binding energy and docking score were used to select the best docking conformations.

Molecular dynamic simulation

In order to assess the best docked complex's binding efficacy and demonstrate how lead compound binding affects a chosen protein's internal dynamics, MD simulation was conducted using the AMBER20 tool [28]. A periodic water box with ff14SB and gaff force fields was used to run the simulation for 100 nanoseconds (ns). Using the Amber 20 tools' Leap module, the topologies of the chosen system were generated [29]. Sodium ions (Na^+) were used to neutralize the system. Pre- and post-simulation processing was carried out using the previously described method [30]. To determine how the top-scoring metabolite stabilized the complex with the target protein, parameters such as RMSD and RMSF were computed. Furthermore, the binding free energy for both protein and ligand were determined through generalized born surface area (MMGB/PBSA) approaches. 1000 snapshots were taken during the trajectory of MD simulation using the same techniques as previously indicated in order to compute the binding free energy differences [31].

Results and discussion

Putative secondary metabolites

AntiSMASH version 6.0 was used for the evaluation of potential BGCs and their secondary metabolites. It was used to predict the gene clusters in *Kytococcus schroeteri* strains H01 and UMB1298. A limited number of 5 potential BGCs were predicted in the UMB1298 strain and 3 potential BGCs in the H01 strain. However, it is worth noting that other related species, such as *K. sedentarius* and *K.*

aerolatus, also exhibit a limited number of BGCs, with only five regions each, according to the AntiSMASH results. This suggests a trend across this genus of having a minimal number of BGCs. The BGCs of *K. schroeteri* include two terpene clusters, one NAPAA cluster, two NI-siderophore clusters, one ribosomally-synthesized RiPP-like post-translationally modified peptide product cluster, and two ectoine clusters (Table 1).

BGC involving NAPAA biosynthesis

According to the *in-silico* analysis, the *K. schroeteri* UMB1298 genome anticipated one NAPAA cluster. It contains up to 21 related genes, including both core and additional biosynthetic genes, which were anticipated to play a part in ϵ -Poly-L-lysine NAPAA (Fig. 1A). This gene cluster is proposed to produce ϵ -Poly-L-lysine NAPAA, whose similarity reached 100 % with the ϵ -Poly-L-lysine biosynthetic gene cluster from *Epichloe festucae* based on the known cluster blast (Fig. 1B). This BGC was present in region 1.1 and spanned over 33.87 kb (Table 1). Pfam domains indicated that more than 35 genes are involved in ϵ -Poly-L-lysine synthesis (Fig. 1C).

BGC involving terpene biosynthesis

This BGC was present in the 1.2 region and spanned 20.89 kb in size (Table 1). It contains at least eighteen related genes, comprising of both core and additional biosynthetic genes, which were anticipated to play a role in the synthesis of carotenoids (Fig. 2A). This cluster was proposed to synthesize carotenoid terpene, and its resemblance reached 50 % with the carotenoid biosynthetic gene cluster from *Dietzia* sp. CQ4 based on the KnownClusterBlast (Fig. 2B). Pfam domains showed that more than 16 genes are involved in carotenoid biosynthesis (Fig. 2C). The next terpene gene cluster of strain H01 was present in region 49.1 and its size spanned 14.33 kb, with 50 % resemblance to the carotenoid BGC from *Dietzia* sp. CQ4 when compared at KnownClusterBlast (Table 1, Fig. S1). This BGC had up to sixteen genes encoding terpene biosynthesis (Fig. S1).

BGC involving ectoine biosynthesis

This BGC was present in region 2.1 and its size spanned 10.41 kb (Table 1). It has up to nine genes, which include both core and additional biosynthetic genes that are involved in ectoine biosynthesis (Fig. 3A). This gene cluster was suggested to synthesize ectoine, and possesses 100 % resemblance with the ectoine biosynthetic gene cluster from *Streptomyces* sp. (Fig. 3B). Pfam domains contain ten genes for ectoine biosynthesis (Fig. 3C). Another ectoine gene cluster of strain H01 was present in region 18.1. Its size spanned 10.41 kb, with 100 % resemblance with the terpene BGC from *Streptomyces* sp. when compared to KnownClusterBlast (Table 1, Fig. S2). This BGC had up to nine genes encoding ectoine biosynthesis (Fig. S2). Ectoine is commonly used in the moisturizing and anti-ageing creams to enhance the skin resistance against the skin cleansers [32].

Table 1
Putative secondary metabolites coding cluster of *K. schroeteri* based on genome analysis by antiSMASH.

Strain Name	Region (Contigs)	Type	Nucleotide Sequence (From-To)	Most Similar Known Cluster	Microorganisms	Similarity Percentage
UMB1298	Region 1.1	NAPAA	27638 –61513	ϵ -Poly-L-lysine	<i>Epichloe festucae</i>	100
	Region 1.2	Terpene	152898 –173788	carotenoid	<i>Dietzia</i> sp. CQ4	50
	Region 2.1	Ectoine	157534 –167950	Ectoine	<i>Streptomyces</i> sp.	100
	Region 5.1	RiPP-like	2156 –12491	-	-	-
	Region 6.1	NI-siderophore	62986 –75466	desferrioxamin B/desferrioxamin E	<i>Streptomyces coelicolor</i> A3(2)	33
H01	Region 18.1	Ectoine	10,242 –20,658	Ectoine	<i>Streptomyces</i> sp.	100
	Region 35.1	NI-siderophore	1 –10,327	desferrioxamin B/desferrioxamin E	<i>Streptomyces coelicolor</i> A3(2)	33
	Region 49.1	Terpene	1906 –16,344	carotenoid	<i>Dietzia</i> sp. CQ4	50



Fig. 1. NAPAA (ϵ -Poly-L-lysine) biosynthetic gene cluster in *K. schroeteri* strain UMB1298. (A) The proposed ϵ -Poly-L-lysine gene cluster in region 1.1 from strain UMB1298 genome: 1, bifunctional (p) ppGpp synthetase/guanosine-3',5'-bis(diphosphate) 3'-pyrophosphohydrolase; 5, cytidylate kinase; 6, amino acid adenylation protein; 10, locus tag (CYJ76_00290) glycine dehydrogenase (aminomethyl-transferring). (B) The comparison between the proposed ϵ -Poly-L-lysine gene cluster (red gene) in strain UMB1298 and the proposed identified cluster of ϵ -Poly-L-lysine in *Epichloe festucae* with a similarity score of 0.38 on the MIBiG website. (C) predicted genes involved in ϵ -Poly-L-lysine biosynthesis based on the Pfam website.

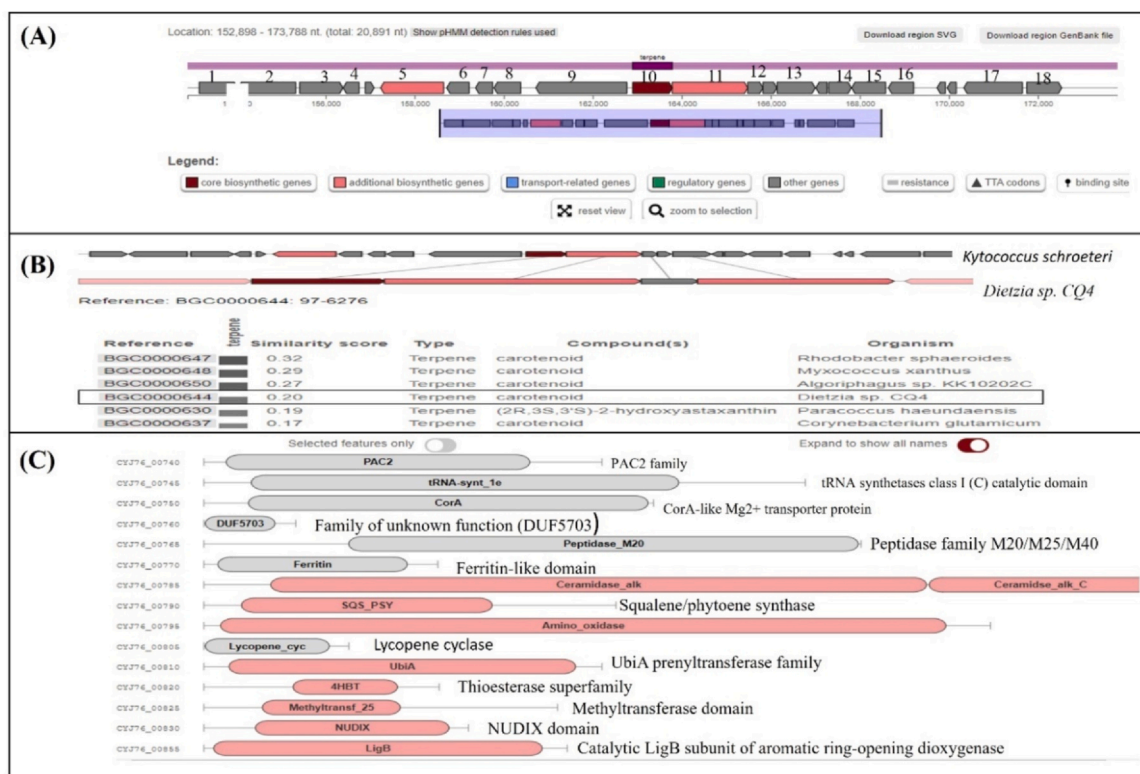


Fig. 2. Terpene (carotenoid) biosynthetic gene cluster in *K. schroeteri* strain UMB1298. (A) The proposed carotenoid gene cluster in region 1.2 from the strain UMB1298 genome: 2, cysteine--1-D-myo-inositol 2-amino-2-deoxy- α -D-glucopyranoside ligase; 6, alkaline ceramidase; 7, phytoene synthase; 8, phytoene desaturase; and 12, isopentenyl-diphosphate delta-isomerase. (B) The comparison between the proposed carotenoid gene cluster in strain UMB1298 and the proposed carotenoid-identified cluster of *Dietzia sp. CQ4* with a similarity score of 0.20 on the MIBiG website. (C) Predicted genes involved in carotenoid biosynthesis based on the Pfam website.

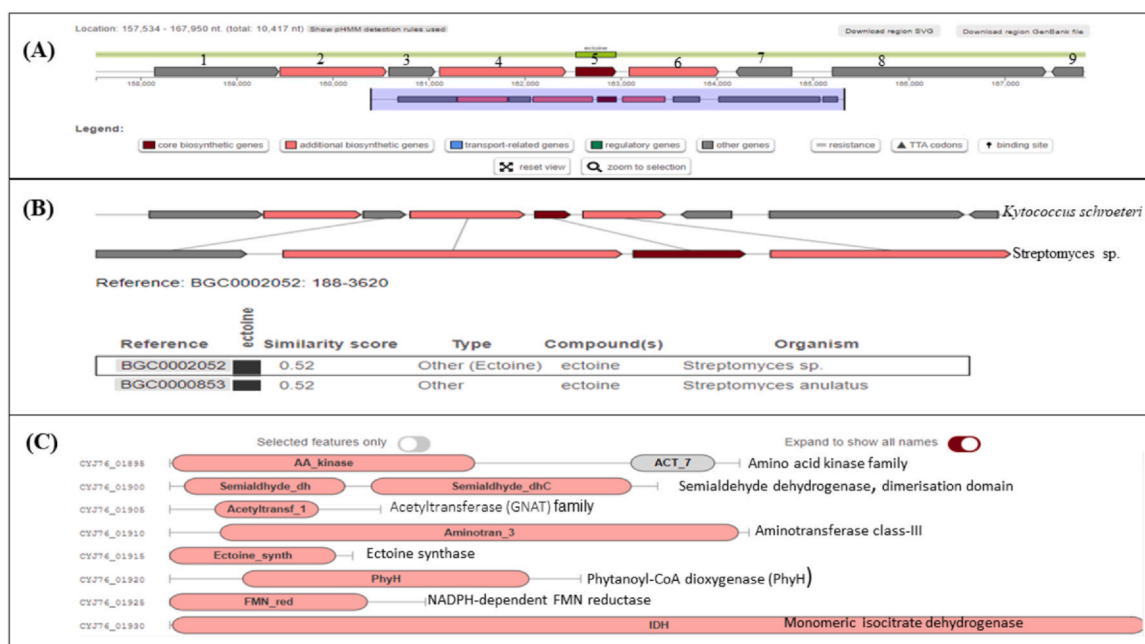


Fig. 3. Ectoine biosynthetic gene cluster in *K. schroeteri* strain UMB1298. (A) The proposed carotenoid gene cluster in region 2.1 from the strain UMB1298 genome: 1, aspartate kinase; 2, aspartate-semialdehyde dehydrogenase; 4, diaminobutyrate-2-oxoglutarate transaminase; and 8, NADP-dependent isocitrate dehydrogenase. (B) The comparison between the proposed ectoine gene cluster in strain UMB1298 and the proposed identified cluster of ectoine in *Streptomyces* sp. with a similarity score reaching 0.52 on the MIBiG website. (C) predicted genes involved in ectoine biosynthesis based on the Pfam website.

BGC involving RiPP biosynthesis

This biosynthetic cluster involves RiPP and was present in region 5.1 and its size spanned 10.33 kb (Table 1). It has over 10 genes that are involved in the core and biosynthetic genes and transport genes, which were proposed to take part in this BGC (Fig. S3). Pfam domains contain nine genes that are involved in the RiPP synthesis (Fig. S3).

BGC involving NI-siderophore biosynthesis

This BGC involves NI-siderophore biosynthesis, whose similarity reached 33 % with the desferrioxamin B BGC from *Streptomyces coelicolor* A3 (2) based on the similarity (Fig. S4B). This BGC was located in the 6.1 region and spanned 12.48 kb in size (Table 1). It has over eight genes, which include both core and additional biosynthetic genes predicted to be involved in siderophore synthesis (Fig. S4A). Pfam domains contain ten genes involved in siderophore synthesis (Fig. S4C). The second NI-siderophore gene cluster of strain H01 existed in region 35.1 and spanned 10.32 kb in size, with 33 % similarity to the NI- siderophore biosynthetic gene cluster from *Streptomyces coelicolor* A3(2) when compared through KnownClusterBlast (Table 1, Fig. S5). This BGC had up to six genes encoding NI-siderophore biosynthesis (Fig. S5). This BGC produced a compound desferrioxamine E, which is reported to exhibit the strongest antibacterial activity against *Yersinia enterocolitica* (Enterobacterales) and *Erwinia amylovora* [33].

Downstream biosynthetic pathways of the BGCs

Region 6.1 contains protein contig with the locus tag CYJ76_04440 involved in the transport of siderophores (Table S1). ABC transporter protein is a subfamily of the bacterial binding protein-dependent transport system family [34]. This region also contains another transport protein of the siderophore (Locus tag: CYJ76_04445). FecCD transporter protein is a subfamily of the bacterial binding protein-dependent transport system family. These Pfams item describes the core components of the multicomponent transport system [35].

Contig 35_11 of region 35.1 and contig CYJ76_04335 of region 6.1 contain the regulatory proteins of siderophore (Table S1 and S2). This regulatory protein from GntR family includes the N-terminal

HTH region of GntR-like bacterial transcription factors. An effector-binding/oligomerization domain is often located at the C-terminus [36]. GntR-like proteins are classified into the following subfamilies: MocR, YtrR, FadR, AraR, HutC, PlmA, DevA, and DasR [37]. Many of these proteins have been experimentally demonstrated to be auto-regulatory, allowing the prediction of operator sites and the finding of cis/trans connections [38]. DasR is a worldwide regulator of primary metabolism and development in *Streptomyces coelicolor*.

These contig also contain another regulatory protein named as the FCD domain (Table S1 and S2). Many GntR family members have this domain as their C-terminal ligand binding domain. This domain binds to a variety of effector chemicals, including lactate, zinc (II), nickel (II), calcium (II), magnesium (II), citrate, sugar acids, sialic acid, and N-acetylglucosamine-6-P, which control gene transcription via the action of the N-terminal DNA-binding domain Pfam:PF00392 [39]. This domain is present in Swiss: P45427 and P31460, which regulate sugar biosynthesis operons. It is also in the known FadR structure, where it binds to acyl-coA; the alpha helical domain [40].

We aim to explore the downstream pathways of Biosynthetic Gene Clusters (BGCs), specifically investigating whether new enzymes are involved in siderophore modification or transport. Our investigation focuses on BGCs linked to transport or regulatory enzymes involved in siderophore transport. The biosynthetic cluster (BGC0001572) predicted by MIBiG website predict putative siderophore MFS transporter DfoS enzymes. The functional annotation of this enzyme was determined by the InterProScan by submitting the amino acid sequence. The InterPro GO terms determined that these are involved in transmembrane transporter function (Fig. 4A) and enable xenobiotic detoxification via transmembrane export across the plasma membrane (Fig. 4B). By looking into these BGCs, we hope to uncover possible mechanisms governing siderophore modification and transport.

The biosynthetic gene cluster (BGC0002464) produces the regulatory enzymes alcaligin siderophore biosynthesis transcriptional regulator AlcR which helps in the regulation of DNA-templated transcription process, functional annotation determined by the InterPro GO terms (Fig. S6). This study aims to identify the unique

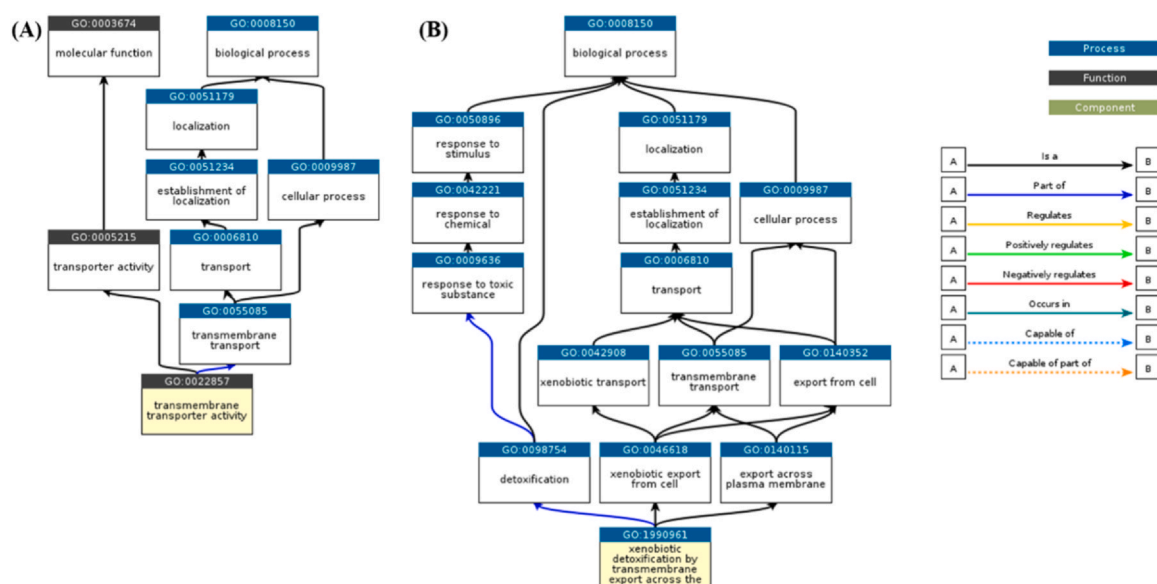


Fig. 4. Functional annotation of siderophores produced. (A) The InterPro GO terms determined that BGC0001572 produces enzymes, that are involved in transmembrane transporter function. (B) represents that siderophores enable the xenobiotic detoxification through transmembrane export across the plasma membrane.

properties and functional diversity of synthetic siderophores by predicting the appearance of novel enzymes translated by BGCs.

Identification of secondary metabolites with drug-like potential

From the antiSMASH results, we found 49 unique secondary metabolites in the *K. schroeteri* bacterial genome (Table S3). The structures of six RiPP, namely SRO15–3108, Triculamin, Alboverticillin, Thatisin, cytolysin ClyL, cytolysin ClyLs, and three NRP, namely ϵ -Poly-L-lysine, γ -Poly-L-2,4-diaminobutyric acid, and rhizomide A, were not found in the available literature. Using in-silico drug discovery techniques, forty putative secondary metabolites identified from the genome mining of these two strains were evaluated for required drug-like characteristics. During this investigation, thirteen metabolites were found to fulfill Lipinski's rules of five (i.e., HBA < 10, HBD < 5, logP < 5, and MW < 500) with zero violations (Table S3). These compounds are likely to exhibit good oral bioavailability in humans. Their molecular properties, including molecular weight and lipophilicity, fall within the suggested limits, making them potential candidates for drug development.

Physicochemical properties

Physicochemical properties such as the lipophilicity (logPO/w), hydrogen bond donors (HBD), hydrogen bond acceptors (HBA), logarithm

of aqueous solubility value (logS), molecular weight, topological polar surface area (TPSA), and molar refractivity (MR) of the thirteen metabolites, following the Lipinski with zero violation, were predicted using the SwissADME web server. The main objective of investigation of these properties was to assess the developed compounds' drug-likeness using important parameters. (Table 2). According to Lipinski's rule of five, all of the top compounds fall into the drug-likeness category, as expected. All of the compounds' expected TPSA values fall between 0 and 140 Å², which is the recognized range. The ideal range of solubility (log S) for the medications was reported to be between 5.0 and 1.0 [41], while the anticipated log S values for the compounds under study were between 5.482 and 1.543. The molecular capacity for hydrophobic interactions with the receptor atoms is linked to the molecular refractivity (MR). All of the top compounds were projected to fall between the recommended range of MR values for drug-like compounds, which is 40 to 130, with the exception of pyrrolomycin, which had a tiny deviance of 39.63 [42].

Pharmacokinetic analyses

To obtain information on the pharmacokinetic properties of putative secondary metabolites, they were subjected to Swiss ADME analysis. The predictions reveal that all the compounds showed high GI absorption (Table S4), indicating their potential for oral drug

Table 2
Physicochemical properties by SwissADME server.

Sr. No.	Compound	Molecular formula	MW (g/mol)	nHA	nHBA	HBD	Log P	Log S	MR	TPSA (Å ²)
1.	Tetrabromopyrrole	C4HBr4N4	382.67	9	0	1	2.23	-5.23	51.59	15.79
2.	2,3,4,5-tetrabromopyrrole	C6H5Br4N	410.73	11	0	0	2.94	-5.42	61.30	4.93
3.	Carquinostatin A	C21H23NO3	337.41	25	3	2	2.80	-4.15	100.78	70.16
4.	Carquinostatin B	C21H23NO4	353.41	26	4	3	2.61	-3.54	101.94	91.39
5.	Ectoine	C6H10N2O2	142.16	10	3	2	0.88	-0.16	44.34	61.39
6.	Trichobrasilenol	C15H26O	222.37	16	1	1	2.95	-3.69	70.42	20.23
7.	Xylarenic acid B	C15H24O3	252.35	18	3	2	2.14	-3.00	72.19	57.53
8.	Brasilane A	C23H39NO6	425.56	30	6	4	2.72	-3.61	114.26	108.25
9.	Brasilane F	C15H26O2	238.37	17	2	2	2.78	-2.94	71.58	20.46
10.	Brasilane E	C23H37NO8	455.54	32	8	4	3.01	-2.64	113.98	137.85
11.	Brasilane D	C23H39NO7	441.56	31	7	5	2.81	-2.84	115.42	128.48
12.	Legonoxamine B	C17H24N2O5	336.38	6	5	3	2.13	-1.66	88.08	106.94
13.	Pyrrolomycin	C4H2Cl2N2O2	180.98	10	2	1	1.01	-2.61	39.63	61.61

nHA represents the number of heteroatoms; nHBA denotes the number of hydrogen bond acceptors; HBD stands for the number of hydrogen bond donors; Log P is the logarithm of the partition coefficient, indicating lipophilicity; Log S is the logarithm of solubility; MR refers to the molar refractivity; and TPSA is the topological polar surface area.

efficacy and optimal bioavailability in systemic circulation. Such compounds can achieve therapeutic concentrations more consistently, enhancing the likelihood of successful treatment outcomes. Phosphorus-glycoprotein binding (also known as binding to the ATP-binding cassette) sucks up the potential for drug efflux from cells. P-gps aid in the transportation of xenobiotics toward blood, urine, and bile and are found in the intestinal epithelium, kidney tubular cells, liver cells, and blood-brain barrier [43]. One noted restriction for CNS-acting medications is binding to P-gps, which suggests potential for intestinal absorption and potential for back-transport toward the blood. Therefore, this property can be favorable in the context of antimicrobial drug development. Carquinostatin B, Brasilane A, Brasilane E, and Brasilane D are suggested to be the P-gp substrates. Being RpfB inhibitors, our top hits may target infections within cells or organs with high P-gp activity (like the lungs, which are a primary site for Tb). As substrates for P-gp, an efflux transporter, these compounds are likely to have decreased intracellular concentrations which can reduce their efficacy. Modulating P-gp activity, for instance through co-administration with P-gp inhibitors like verapamil or ketoconazole, can increase the bioavailability of these drugs [44]. Knowing that these inhibitors are P-gp substrates can guide the development of novel drug formulations that enhance their permeability and stability. For example, the use of nanoparticles or liposomal delivery systems can protect drugs from P-gp-mediated efflux, as has been demonstrated with other drugs [45].

The results of BBB permeability investigations identified that only the compounds Carquinostatin B, Ectoin, Brasilane A, Brasilane E, Brasilane D, and Legonoxamine B were unable to penetrate the BBB (Table S4). All other top hits are BBB permeable. Antimicrobial agents that are substrates for P-gp and have a reduced propensity to cross the BBB are less likely to accumulate in the brain. This can potentially reduce the risk of neurotoxicity and other CNS-related side effects. Antibiotics that cross the BBB can lead to unwanted exposure in the central nervous system (CNS), which might result in neurotoxicity or other adverse neurological effects. Developing antibiotics with molecular structures that inherently limit their ability to cross the BBB can prevent unintended CNS exposure. Altering properties such as molecular size, polarity, and charge can effectively reduce BBB permeability. Targeted drug delivery systems could be employed to ensure that antibiotics are delivered only to intended sites of infection, thus minimizing exposure to the CNS [46].

The potential for drug-drug interactions is linked to CYP isoenzyme inhibition. Among these isoenzymes, CYP2D6, and CYP3A4 are primarily responsible for metabolism [47]. The results (Table S4) for CYP3A4 and CYP2D6 indicated that all compounds are non-inhibitors except Carquinostatin A, which is predicted to be the inhibitor for CYP3A4. These latest findings imply that these substances wouldn't interact with one another due to inhibition of the cytochromes, which metabolize a large range of medications, and hence can be administered safely in combination with other drugs.

Impact of CYP inhibition on TB combination therapy

When considering combination therapy with established antibiotics like isoniazid, rifampin, pyrazinamide, ethambutol, or streptomycin, it's crucial to consider the potential cytochrome P450 (CYP) inhibition by our top hits. Certain CYP enzymes metabolize these antibiotics, and inhibition by our compounds could impact the antibiotics' effectiveness and safety.

Among the top hits, Tetra-bromopyrrole, 2,3,4,5-tetra-bromopyrrole, Pyrrolomycin, Carquinostatin A, and Carquinostatin B inhibit CYP1A2; the first two also inhibit CYP2C9. Additionally, Trichobrasilenol inhibits CYP2C9, and notably, Carquinostatin A inhibits multiple CYPs, including CYP1A2, CYP2C19, CYP2C9, and CYP3A4 (Table S4). This broad spectrum of CYP inhibition poses

significant concerns for combination therapy, particularly with antibiotics metabolized by these enzymes.

According to ADMETlab 2.0 data, isoniazid and streptomycin, which are not metabolized by CYP enzymes, are safer options for combination therapy with our top hits. However, rifampin, a substrate of CYP2C19 and CYP3A4, could have its efficacy compromised when combined with Carquinostatin A. Likewise, pyrazinamide, metabolized by CYP1A2, may be adversely affected when used with our top hits that inhibit this enzyme. Thus, careful selection and monitoring of antibiotic combinations are imperative to avoid harmful drug-drug interactions and ensure therapeutic efficacy.

Toxicity and target prediction analysis

The toxicity profile of the compounds was identified and compared with the reference compound (Table S5). The results show that, like the reference, all compounds are non-hepatotoxic and non-cytotoxic except pyrrolomycin, which was predicted to be hepatotoxic. 2,3,4,5-tetrabromopyrrole has one of the lowest LD50 values at 313 mg/kg, followed closely by pyrrolomycin with 435 mg/kg and carboquinostatin A with 136 mg/kg. These values suggest that these compounds are comparatively more toxic than the other top compounds. Legonoxamine B, Ectoine, and Tetrabromopyrrole have moderate LD50 values of 990 mg/kg, 1000 mg/kg, and 1000 mg/kg, respectively, implying that they are of intermediate toxicity. Compounds like brasilane E, xylarenic acid B, and brasilane F demonstrate high LD50 values ranging from 2266 mg/kg to 5000 mg/kg, implying lower acute toxicity. Notably, Brasilane A, Brasilane D, and Trichobrasilenol have the highest LD50 values at 5000 mg/kg, categorizing them as the least acutely toxic among the group.

Target prediction analysis was carried out to evaluate potential interactions between the top compounds and human proteins. This was done to see if these compounds might accidentally target and inhibit human proteins. Interactions are quantified in the results as probability values that range from 0 to 1. A score closer to 1 denotes a higher likelihood that the molecule is targeting a particular human protein. The metabolites were examined by Swiss target prediction server. The results of all compounds were in the range of 0 to 0.51, predicting that human proteins have the least possible chances of being inhibited (Table 3). Targets of pyrrolomycin and 2,3,4,5-tetrabromopyrrole show that they have no probability of inhibiting human proteins, which is relevant considering that these compounds should not affect human protein function and therefore can be proposed as drug candidates. Neither brasilane A nor tetrabromopyrrole were predicted to target humans (Table 3).

Validation of docking protocol

We compared the co-crystallized benzamidine-RpfB complex (PDB ID: 4EMN) with the re-docked form of the same complex to validate our docking methodology. These structures were superimposed, and we found that they had similar binding sites (Fig. 5A), which suggests that the ligands interact similarly in both cases. With a noticeably low RMSD value of 0.96, the sequence alignment further supported this finding by highlighting the similar ligand interactions in both sets.

Molecular docking inspection of putative secondary metabolites against RpfB target

Docking analysis is performed mainly to determine the protein-ligand interactions. Thirteen metabolites with druggability potential based on the Lipinski rule, along with the reference compounds, were subjected to docking analysis against the RpfB protein of Mtb. Out of thirteen metabolites, nine compounds showed the highest docking score compared to the reference compound, i.e.,

Table 3
Target fishing analysis by Swiss Target Prediction.

Predicted Targets	Probability
Targets of Pyrrolomycin	
Vascular cell adhesion protein 1	0.0
Nitric-oxide synthase, brain	0.0
Nitric oxide synthase, inducible	0.0
Acyl coenzyme A: cholesterol acyltransferase	0.0
Signal transducer and activator of transcription 3	0.0
Targets of 2,3,4,5-tetrabromopyrrole	
Androgen Receptor	0.0
Targets of Carquinostatin A	
Thrombin and coagulation factor X	0.09
Sulfonylurea receptor 2	0.09
Vasopressin V1a receptor	0.09
Oxytocin receptor	0.09
MAP kinase p38 alpha	0.09
Targets of Carquinostatin B	
Hypoxia-inducible factor 1 alpha	0.10
Alkaline phosphatase, tissue-nonspecific isozyme	0.10
Anandamide amidohydrolase	0.10
Poly [ADP-ribose] polymerase-1	0.10
Phosphodiesterase 10 A	0.10
Targets of Ectoine	
GABA-A receptor; alpha-1/beta-2/ gamma-2	0.07
GABA A receptor alpha-3/beta-2/ gamma-2	0.07
GABA A receptor alpha-2/beta-2/ gamma-2	0.07
GABA receptor rho-1 subunit	0.07
Dopamine transporter (by homology)	0.0
Targets of Trichobrasilenol	
UDP-glucuronosyltransferase 2B7	0.51
Testis-specific androgen-binding protein	0.06
Androgen Receptor (by homology)	0.06
Cytochrome P450 19A1	0.06
Dual specificity phosphatase Cdc25 A	0.06
Targets of Bylarenic acid B	
DNA polymerase beta (by homology)	0.09
Potassium-transporting ATPase alpha chain 2	0.08
UDP-glucuronosyltransferase 2B7	0.06
Protein farnesyltransferase	0.06
11-beta-hydroxysteroid dehydrogenase 1	0.05
Targets of Brasilane F	
Androgen Receptor	0.17
Testis-specific androgen-binding protein	0.08
UDP-glucuronosyltransferase 2B7	0.06
Cytochrome P450 19A1	0.05
Estrogen receptor alpha	0.05
Targets of Brasilane E	
Myelin-associated glycoprotein	0.12
Galectin-3	0.10
HERG	0.10
GABA receptor alpha-5 subunit	0.10
Adenosine kinase	0.10
Targets of Brasilane D	
Sodium/glucose cotransporter 2	0.11
Sodium/glucose cotransporter 1	0.11
Dual specificity mitogen-activated protein kinase kinase 1	0.11
Adenosine A1 receptor	0.11
Adenosine A3 receptor	0.11
Targets of Legonoxamine B	
Integrin alpha-IIb/ beta-3	0.10
Matrix metalloproteinase 1	0.10
Neprilysin (by homology)	0.10
Prostanoid IP receptor	0.10
Phosphodiesterase 5 A	0.10

benzamidine (Table 4). The docking scores of these compounds range from -7.12 to -4.78 kJ/mol. Legonoxamine B, which showed the highest docking score of -7.12 kJ/mol, was classified as the other BGC (cluster that does not fall into any other category and contains a secondary metabolite-related protein). Brasilane A, F, E, and D are from the terpene biosynthetic gene cluster and showed docking scores of -6.76 , -5.17 , -5.45 , and -6.56 KJ/mol, respectively. Carquinostatin A and Carquinostatin B are alkaloids in nature and have docking scores of -6.12 and -5.19 kJ/mol, respectively. Trichobrasilenol and Xylarenic acid B are also terpenes with docking scores

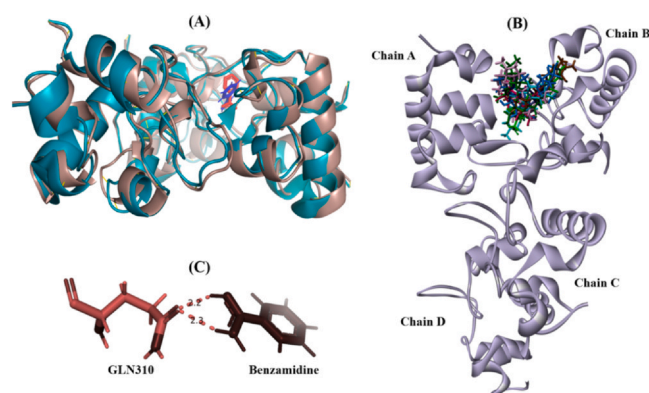


Fig. 5. (A) Superimposition of binding location of benzamidine (red) from crystal structure of RpfB (PDB ID: 4EMN) (dark salmon) and binding location of our re-docked benzamidine (blue) and RpfB (teal) complex showing binding of benzamidine at same location. (B) Binding of top compounds in the catalytic pocket of RpfB. The catalytic pocket resides between chains A and B: legonoxamine B (green), trichobrasilenol (blue), brasilane D (zinc-blue), brasilane F (dark pink), brasilane E (army green), carquinostatin B (brown), carquinostatin A (cyan), xylarenic acid B (purple pink), and brasilane A (light pink). (C) Molecular interaction (red) of benzamidine (dark red) from the re-docked benzamidine-RpfB complex.

of -5.06 and -4.78 kJ/mol, respectively. Conclusively, the top hits had significantly better bonding with the RpfB of Mtb compared to the reference (Table 4). Notably, most of the secondary metabolites were found to be terpene in nature. It has been shown that terpenes have strong antibacterial activity, acting both bacteriostatically and bactericidally against tested microorganisms [48]. Monoterpenes such as carvacrol, menthol, thymol, and geraniol are effective against both Gram-positive and Gram-negative bacteria. Geraniol was claimed to have medicinal potential against multi-drug-resistant *Enterobacter aerogenes* [49]. Two triterpenes, namely bonianic acid A and B, were particularly active for Mtb [50]. Additionally, in clinical settings, β -sitosterol has demonstrated immune-enhancing ability in TB patients [51].

Protein-ligand interactions

The interactions of ligands within the receptor-binding cleft are the essential results after docking [24]. The molecular interactions of each docked complex were analyzed to understand the stability of the complexes. The results of docking indicated that all the top compounds have almost the same interactions with the amino acid residues such as GLU292, ASN303, GLN310, and ASP312 in the receptor binding pocket, along with the hydrophobic interactions, i.e., CYS291, PHE311, ASP312, GLN310, GLN347, VAL309, THR315, TRP352, PRO353, VAL354, and CYS355 (Table 4). Among the active site residues of RpfB, GLU292 is central to the catalytic activity, strictly conserved in the active site, and responsible for its metabolic functions [52]. A hydrophobic environment that surrounds the GLU292 is crucial for the catalytic ability of RpfB protein. This environment probably enhances the chance of hydrogen-bonding interaction with the RpfB inhibitors [52].

In our results, the top nine compounds are bound in the catalytic pocket between chains A and B (Fig. 5B), showing interactions with the significant amino acid residues. The H-bond interactions of the nine compounds with GLU292 were mainly observed (Table 4). Important interactions are seen between the N-acetyl-acetylamine moiety of the Mtb cell wall and Glu292 and GLN310 of RpfB. The process that promotes cell wall integrity preservation, growth, and division is aided by this interaction (between RpfB and Mtb cell walls) [52]. Therefore, targeting these crucial residues significantly inhibits RpfB activity. The interactions of the legonoxamine B moiety show that it formed three hydrogen bonds: one bond with GLU292

Table 4
Interaction profile of standard and top-scoring metabolites with RpfB (PDB ID-4EMN) protein.

Sr.no	Compound name	Docking score (KJ/mol)	Hydrogen bond interaction	Hydrophobic interactions
Ref.	Benzamidine	-4.14	GLN310	CYS291, GLU292, ASN296, THR301, ASN303, TYR305, ALA293, ILE299,
1.	Carquinostatin A	-6.12	GLU292	CYS291, ILE299, ASN296, ASN303, ALA293, GLY308, ALA351, TRP352, PRO353, VAL354
2.	Carquinostatin B	-5.19	ASP312	VAL309, ASN303, GLN310, GLY350, ALA351, THR315, TRP352, PRO353, VAL354, PHE311
3.	Legonoxamine B	-7.12	GLU292, ASN303	CYS291, ALA293, ILE299, GLY294, THR301, TYR305, PHE311, ASP312, GLN310, GLN347, VAL309, THR315, TRP352, PRO353, VAL354, CYS355
4.	Trichobrasilenol	-5.06	GLU292, ASN303	CYS291, ALA293, ASN303, GLN310, TYR305, ALA351, TRP352, PRO353, VAL354, CYS355
5.	Xylarenic acid B	-4.78	ASN303	CYS291, ALA293, GLU292, ASN303, TYR305, GLY350, TRP352, PRO353, VAL354, CYS355
6.	Brasilane A	-6.56	GLU292, ASN303	GLY350, ALA351, TRP352, PRO353, VAL354, CYS355, CYS291, ALA293, TRP285, ILE299, TYR305
7.	Brasilane F	-5.17	GLU292, ASN303	CYS291, GLY304, TYR305, ILE299, ALA293, GLN310, ALA351, TRP352, PRO353, VAL354, CYS355
8.	Brasilane E	-5.45	GLU292(2 bonds), ASN303	CYS291, ALA293, ILE299, ALA295, GLN310, THR301, ALA351, TRP352, PRO353, VAL354, CYS355
9.	Brasilane D	-6.76	ASN303, GLN310	CYS291, GLY304, VAL309, GLY350, ALA351, TRP352, PRO353, VAL354, GLN347

of chain A, a second with GLU292 of chain B, and a third with the ASN303 of chain B of protein RpfB (Fig. 6A,B). Additionally, hydrophobic interactions with amino acids such as CYS291, THR301, PHE311, ASP312, GLN347, VAL309, THR315, TRP352, PRO353, and VAL354 were observed (Fig. 6C, Table 4).

GLU292 serves as a critical proton donor to the glycosidic bond of the substrate in the catalytic mechanism of RpfB. By interacting with GLU292, legonoxamine B could prevent this proton donation, thereby blocking the cleavage of the glycosidic bond essential for peptidoglycan breakdown. Moreover, the action of GLU292 is central to the formation of an oxocarbenium ion during the catalytic cycle of RpfB. Inhibiting this process would prevent the destabilization of the substrate necessary for its breakdown, effectively stopping the resuscitation of dormant Mtb. Lastly, a water molecule (W1) tightly bound and influenced by GLU292 plays a crucial role in catalysis [53]. Legonoxamine B binding with GLU292 could disrupt this water-mediated process, which is essential for the proper positioning and activation of GLU292 for substrate interaction. The RMSD, RMSF and MMGB/PBSA analysis further highlighted the high binding affinity of legonoxamine B for the RpfB (Fig. 10). Therefore, targeting GLU292 in RpfB by legonoxamine B and most of the top inhibitors could effectively inhibit a critical step in the pathway that allows Mtb to transition from a dormant to an active state, providing a novel approach to control and potentially eliminate Tb infections.

The standard treatment for drug-susceptible Tb includes a combination of at least four antimicrobial agents administered over a 6-month period. Key drugs include isoniazid, rifampin, pyrazinamide, and ethambutol or streptomycin [54]. It is suggested that combining RpfB inhibitors (such as legonoxamine B and other top hits) with conventional antibiotics could potentially restore sensitivity in drug-resistant bacteria and enhance the antibacterial activity, similar to the efflux pump inhibitors [55].

Trichobrasilenol displayed the formation of two H-bonds with the residues GLU292 of chain A and ASN303 of chain B (Fig. 6D,E). Hydrophobic interactions of Trichobrasilenol are observed with CYS291, ALA293, ASN303, GLN310, TYR305, ALA351, TRP352, PRO353, VAL354, and CYS355 (Fig. 6F, Table 4).

In the case of brasilane D, two hydrogen bonds were formed with GLU292, GLN310 of chain A and one bond with ASN303 of chain B (Fig. 7A,B); and the residues, namely CYS291, GLY304, VAL309, GLY350, ALA351, TRP352, PRO353, VAL354, and GLN347, had hydrophobic interactions with the moiety (Fig. 7C, Table 4). Brasilane D was the only top compound that exhibited covalent bonding with the GLN310 similar to benzamidine (Fig. 5 C). Bonding with Brasilane E exhibited three hydrogen bonds, two bonds with the same residue, i.e., GLU292 of chain A, and one with ASN303 of chain B (Fig. 7D,E). Furthermore, the hydrophobic interactions with the RpfB protein involve CYS291, ALA293, ILE299, ALA295, GLN310, THR301, ALA351, TRP352, PRO353, VAL354, and CYS355 residues (Fig. 7F).

Both brasilane A and brasilane F formed two hydrogen bonds with GLU292 of chain A and ASN303 of chain B in the catalytic binding site (Fig. 8), along with the akin hydrophobic amino acids (Table 4). Carquinostatin A, one of the top hits, demonstrated hydrogen bonding to chain A's GLU292 (Fig. 9A,B), indicating a physiologically meaningful binding in the suppression of RpfB. The hydrophobic interactions of Carquinostatin A include residues i.e., CYS291, ILE299, ASN296, ASN303, ALA293, GLY308, ALA351, TRP352, PRO353, and VAL354 (Fig. 9C). In the catalytic site of RpfB, the carquinostatin B moiety and ASP312 of chain A bonded by hydrogen bonding (Fig. 9D,E). The larger negatively charged electrostatic potential in the RpfB substrate binding cleft is caused by ASP312 and could be linked to increased enzymatic activity [52]. The hydrophobic interactions of Carquinostatin B with the protein involve VAL309, ASN303, GLN310, GLY350, ALA351, THR315, TRP352, PRO353, VAL354, and PHE311 (Fig. 9F, Table 4). The xylarenic acid B moiety shows one hydrogen bond with ASN303 of chain A with significant hydrophobic interactions (Fig. S7, Table 4).

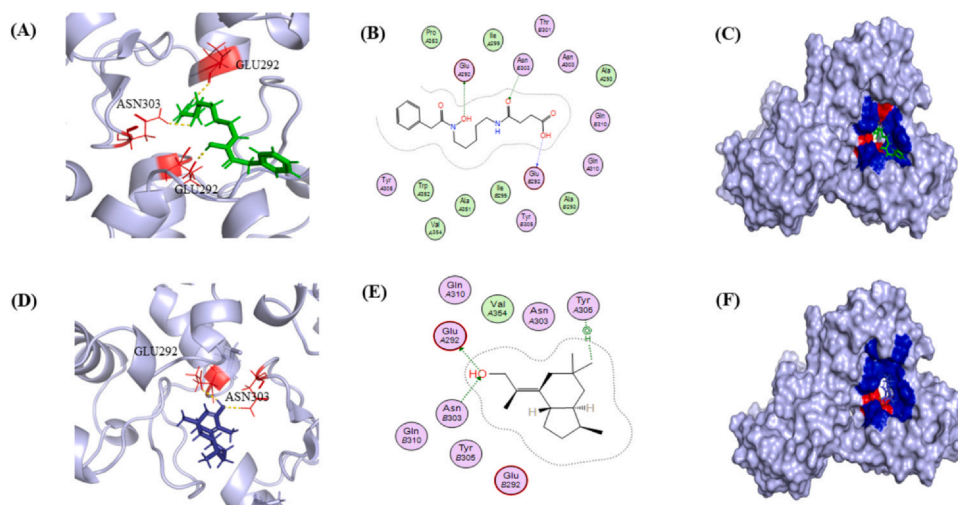


Fig. 6. Molecular interactions of protein (light blue) with legonoxamine B (green) and trichobrasilenol (blue). (A) and (D) represent their 3D-interacting residues (red). (B) and (E) represent their 2D schematic interactions. Lastly, (C) and (F) represent the nature of interactions depicted by hydrophilic (red) and hydrophobic (blue) interactions.

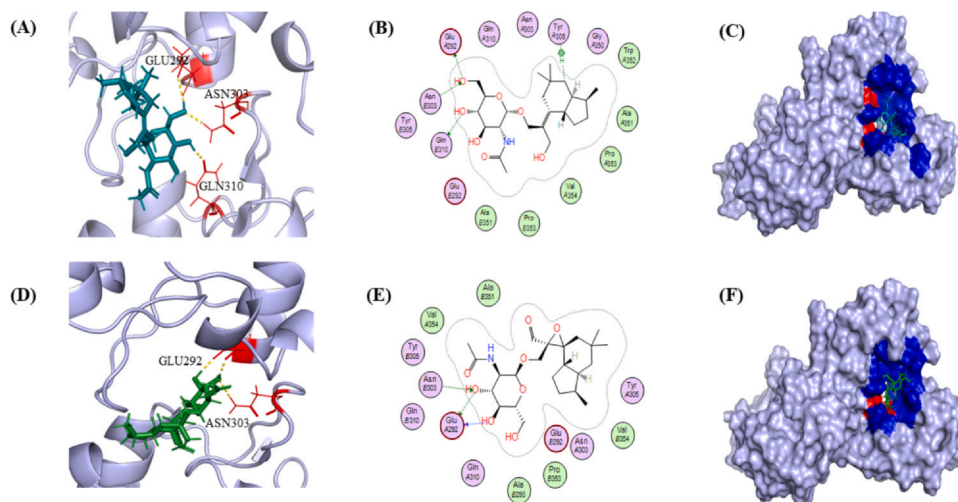


Fig. 7. Molecular interactions of protein (light blue) with Brasilane D (zinc-blue) and Brasilane E (army green). (A) and (D) represent their 3D-interacting residues (red). (B) and (E) represent their 2D schematic interactions. Lastly, (C) and (F) represent the nature of interactions depicted by hydrophilic (red) and hydrophobic (blue) interactions.

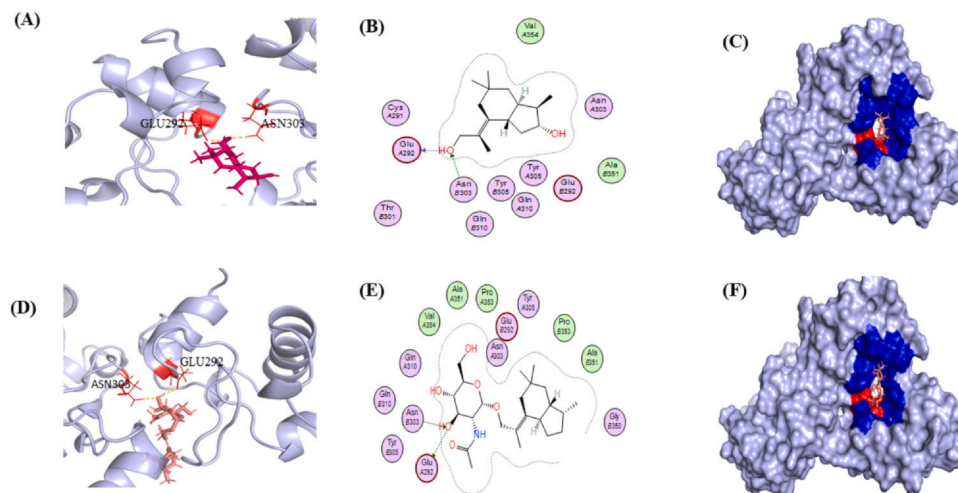


Fig. 8. Molecular interactions of protein (light blue) with Brasilane F (dark pink) and Brasilane A (light pink). (A) and (D) represent their 3D-interacting residues (red). (B) and (E) represent their 2D schematic interactions. Lastly, (C) and (F) represent the nature of interactions depicted by hydrophilic (red) and hydrophobic (blue) interactions.

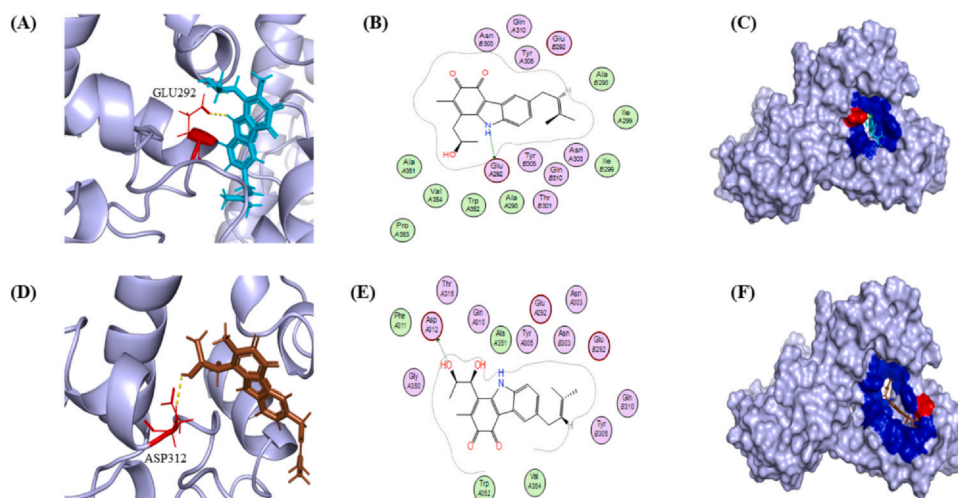


Fig. 9. Molecular interactions of the protein (light blue) with Carquinostatin A (cyan) and Carquinostatin B (brown). (A) and (D) represent their 3D-interacting residues (red). (B) and (E) represent their 2D schematic interactions. Lastly, (C) and (F) represent the nature of interactions depicted by hydrophilic (red) and hydrophobic (blue) interactions.

MD simulation analysis of legonoxamine B and RpfB complex

In order to validate the interactions analysis of the lead compounds with the receptor protein, one compound, i.e., the Legonoxamine B-RpfB complex, was subjected to molecular dynamics simulation analysis. QtGrace was used to construct the root mean square deviation (RMSD) plots for the carbon atoms of the protein and the ligand. The RMSD plot shows that at around 30 ns from the start of the simulation, the system reached equilibration with low fluctuations, indicating that the selected metabolite was stable and remained in the active site of the protein until the end with an average RMSD value of 5.97 Å (Fig. 10A). Fig. 10B depicts the RMSD plot for Legonoxamine B, where a stable behavior with an average RMSD value of 3.03 Å was observed. To take into account the variability of amino acid residues, the RMSF of the backbone residue of RpfB was determined. Understanding the degree to which ligand

binding influences protein flexibility during the simulation run required a closer look at each residue's flexibility. Fig. 10C shows the estimated RMSF value for the chosen complex. The RMSF plot highlights that the binding of the selected compound caused no major changes or noteworthy fluctuations in the structure of the protein. Hence, it is concluded that the selected compound exhibited the greatest affinity for the studied protein and could be a potent antibacterial against antibiotic resistance. AMBER20's MMPBSA.py module was used to compute the binding free energy of selected top-scored docked complexes. The findings of the MMGB/PBSA analysis are summarized in Table 5; both approaches revealed the formation of very stable complex relationships. The entropy term is excluded because it cannot be quantified in some instances due to convergence concerns. Higher values of negative binding free energy suggested a more robust connection and higher affinity between the small compounds and the protein. The calculated binding free

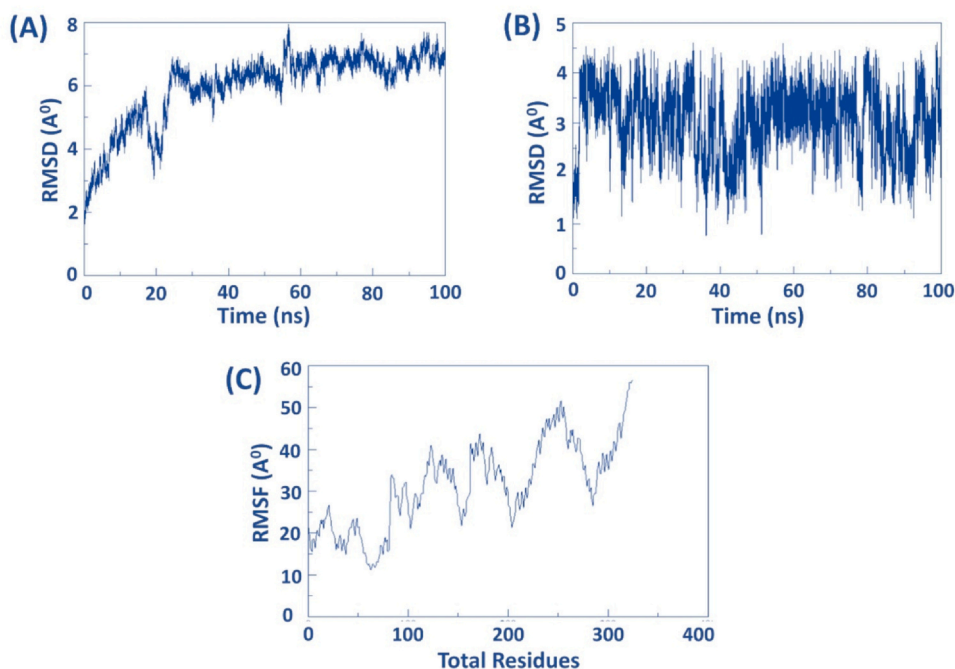


Fig. 10. Molecular dynamic simulation analysis of Legonoxamine B (A) Backbone root mean square deviation (RMSD); (B) C α RMSD (C) Root mean square fluctuation (RMSF).

Table 5

Binding energy values for top-scored docked complex. The units of energy are kcal/mol.

Energy Components	MMGBSA		MMPBSAs	
	Average	Std. Dev.	Average	Std. Dev.
Van der Waal energy (VDWAALS)	-17.4948	2.8018	-17.4948	2.8018
Electrostatic (EEL)	-15.6415	7.5385	-15.6415	7.5385
EGB	25.6119	6.7770	EPB 25.9193	7.7835
ESURF	-2.6951	0.3950	ENPOLAR -2.4184	0.2532
DELTA G gas	-33.1363	8.0357	-33.1363	8.0357
DELTA G solv	22.9168	6.5892	23.5009	7.6556
DELTA TOTAL	-10.2195	2.8771	-9.6354	3.0600

energy of docked Legonoxamine B and RpfB after a 100 ns MD simulation was negative, indicating that Legonoxamine B exhibited a high binding affinity for RpfB.

Conclusion

In this study, we have uncovered the potential of *Kytococcus schroeteri* strains for producing secondary metabolites that show significant promise as therapeutic candidates through genome mining. The key findings from our study highlight the drug-like properties and good pharmacokinetic profiles of these metabolites as antibacterial drugs, particularly the thirteen that precisely follow Lipinski's rule of five. This compliance suggests a decreased risk of negative side effects, off-target effects, and drug-drug interactions. Compounds such as trichobrasilen, xylarenic acid B, and a series of brasilanes exhibit impressive safety profiles, highlighting their low acute toxicity. Additionally, their higher docking scores over the standard compound against the important drug target, i.e., RpfB of *Mycobacterium tuberculosis*, underscore their potential efficacy. The specific interaction of several metabolites, such as carquinostatin A, Legonoxamine B, trichobrasileno, and brasilanes A, E, and F, with the catalytically significant residue GLU292 of the RpfB protein cannot be understated. The MD simulation results from one of the complexes further validated the binding affinity of the top compounds with the target enzyme. Our study highlighted the potential of secondary metabolites predicted from the genome of *Kytococcus schroeteri* as novel antibacterial agents. Their favorable *in-silico* profiles provide a solid framework for further investigation and experimental studies that may usher in a new era in the fight against antibiotic resistance.

Author Contributions

Conceptualization, M.S., A.A.M.; methodology, F.K., I.A., and S.W.A.; formal analysis, F.K., D.L., S.W.A., S.S., and R.U.; investigation, F.K., I.A., and U.N.; data curation, M.A.I.; writing—original draft preparation, F.K., D.L., I.A., S.W.A., and R.U.; writing—review and editing, M.S., A.A.M., and W.L.; supervision, M.S.; project administration, M.S.; funding acquisition, M.A.I.; All authors have read and agreed to the published version of the manuscript.

Institutional review board statement

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Informed consent statement

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Data availability

All datasets generated or analyzed in this study are included in this article and its [supplementary information files](#).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests to declare.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jiph.2024.102598](https://doi.org/10.1016/j.jiph.2024.102598).

References

- [1] Bagelman S, Zvigule-Neidere G. Insight into kytococcus schroeteri infection management: a case report and review. *Infect Dis Rep* 2021;13:230–8. <https://doi.org/10.3390/IDR13010026>
- [2] Yun J, Kim TS, Park H, Park JH. A case of infective endocarditis caused by kytococcus schroeteri. *Lab Med Online* 2021;11:69–72. <https://doi.org/10.47429/lmo.2021.11.1.69>
- [3] Narayanan Z, Glick BR. Secondary metabolites produced by plant growth-promoting bacterial endophytes. *Microorganisms* 2022;10:2008. <https://doi.org/10.3390/microorganisms10102008>
- [4] Boukerb AM, Robert M, Teteneva NA, Danilova ND, Zhurina MV, Mart'yanov SV, et al. Draft genome sequence of kytococcus schroeteri strain H01, isolated from human skin. *Microbiol Resour Announc* 2019;8. <https://doi.org/10.1128/MRA.01081-19>
- [5] Bagelman S, Zvigule-Neidere G. Insight into kytococcus schroeteri infection management: a case report and review. *Infect Dis Rep* 2021;13:230–8. <https://doi.org/10.3390/IDR13010026>
- [6] Xiong Z, Wang R, Xia T, Zhang S, Ma S, Guo Z. Natural products and biological activity from actinomycetes associated with marine algae. *Molecules* 2023;28:5138. <https://doi.org/10.3390/molecules28135138>
- [7] Aliero A, Emmanue E, Josephat M, Aliyu S, Okech M, Odda J. Antibacterial activity of actinomycetes isolated from waste dump soil from Western Uganda. *Microbiol Res J Int* 2017;21:1–14. <https://doi.org/10.9734/MRJI/2017/36459>
- [8] WHO. Treatment of drug-susceptible tuberculosis: rapid communication. World Health Organization 2021;5. (<https://www.who.int/publications/i/item/9789240028678>) (accessed March 10, 2023).
- [9] Rabaan AA, Garout M, Aljeldah M, Al Shammari BR, Alawfi A, Alshengeti A, et al. Anti-tubercular activity evaluation of natural compounds by targeting *Mycobacterium tuberculosis* resuscitation promoting factor B inhibition: An *in silico* study. *Mol Divers* 2023. <https://doi.org/10.1007/s11030-023-10632-8>
- [10] Chouhan M, Tiwari PK, Moustafa M, Chaubey KK, Gupta A, Kumar R, et al. Inhibition of *Mycobacterium tuberculosis* resuscitation-promoting factor B (RpfB) by microbially derived natural compounds: a computational study. *J Biomol Struct Dyn* 2023;1–12. <https://doi.org/10.1080/07391102.2023.2208214>
- [11] Amin DH, Abolmaaty A, Borsetto C, Tolba S, Abdallah NA, Wellington EMH. *In silico* genomic mining reveals unexplored bioactive potential of rare actinobacteria isolated from Egyptian soil. *Bull Natl Res Cent* 2019;43:78. <https://doi.org/10.1186/s42269-019-0121-y>
- [12] Shah M, Gul S, Amjad A, Sameem Javed M, Fatima B, Nawaz H, et al. Genome mining of streptomyces formicae KY5 for potential drug like natural products characterizations. *J Proteom Bioinform* 2019;12. <https://doi.org/10.35248/0974-276X.19.12.505>
- [13] Loureiro C, Medema MH, van der Oost J, Sipkema D. Exploration and exploitation of the environment for novel specialized metabolites. *Curr Opin Biotechnol* 2018;50:206–13. <https://doi.org/10.1016/j.copbio.2018.01.017>
- [14] Rehan M, Gueddou A, Alharbi A, Ben Abdelmalek I. *In silico* prediction of secondary metabolites and biosynthetic gene clusters analysis of streptomyces thinghirensis HM3 isolated from arid soil. *Fermentation* 2023;9:65. <https://doi.org/10.3390/fermentation9010065>

- [15] Shehroz M, Aslam M, Ali Khan M, Aiman S, Gul Afridi S, Khan A. The in silico characterization of a salicylic acid analogue coding gene clusters in selected *Pseudomonas fluorescens* strains. Iran J Biotechnol 2019;17:e2250. <https://doi.org/10.30498/IJB.2019.95299>
- [16] Evans DA. History of the harvard chemdraw project. Angew Chem - Int Ed 2014;53:11140–5. <https://doi.org/10.1002/anie.201405820>
- [17] Alananzeh WA, Al-qattan MN, Ayipo YO, Mordi MN. N-substituted tetrahydro-beta-carboline as mu-opioid receptors ligands: in silico study; molecular docking, ADMET and molecular dynamics approach. Mol Divers 2023. <https://doi.org/10.1007/s11030-023-10655-1>
- [18] Wadapurkar RM, Shilpa MD, Katti AKS, Sulochana MB. In silico drug design for staphylococcus aureus and development of host-pathogen interaction network. Inf Med Unlocked 2018;10:58–70. <https://doi.org/10.1016/j.imu.2017.11.002>
- [19] Daina A, Michielin O, Zoete V. SwissADME: a free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules. Sci Rep 2017;7:42717. <https://doi.org/10.1038/srep42717>
- [20] Amorim JC, Cabrera Bermeo AE, Vásquez Urgilés VE, Martínez León MR, Carpio Arévalo JM. An In-Silico Evaluation of Anthraquinones as Potential Inhibitors of DNA Gyrase B of *Mycobacterium tuberculosis*. Microorganisms 2022;10:2434. <https://doi.org/10.3390/microorganisms10122434>
- [21] Banerjee P, Eckert AO, Schrey AK, Preissner R. ProTox-II: a webserver for the prediction of toxicity of chemicals. Nucleic Acids Res 2018;46:W257–63. <https://doi.org/10.1093/nar/gky318>
- [22] Daina A, Michielin O, Zoete V. SwissTargetPrediction: updated data and new features for efficient prediction of protein targets of small molecules. Nucleic Acids Res 2019;47:W357–64. <https://doi.org/10.1093/nar/gkz382>
- [23] Shah M, Khan F, Ahmad I, Deng C-L, Perveen A, Iqbal A, et al. Computer-aided identification of *Mycobacterium tuberculosis* resuscitation-promoting factor B (RpfB) inhibitors from *Gymnema sylvestre* natural products. Front Pharm 2023;14. <https://doi.org/10.3389/fphar.2023.1325227>
- [24] Shah M, Yamin R, Ahmad I, Wu G, Jahangir Z, Shamim A, et al. In-silico evaluation of natural alkaloids against the main protease and spike glycoprotein as potential therapeutic agents for SARS-CoV-2. PLoS One 2024;19:e0294769. <https://doi.org/10.1371/journal.pone.0294769>
- [25] Ahmad I, Khalid H, Perveen A, Shehroz M, Nishan U, Rahman FU, et al. Identification of novel quinolone and quinazoline alkaloids as phosphodiesterase 10A inhibitors for parkinson's disease through a computational approach. ACS Omega 2024;9:16262–78. <https://doi.org/10.1021/acsomega.3c10351>
- [26] Shah M, Bashir S, Jaan S, Nawaz H, Nishan U, Abbasi SW, et al. Computational analysis of plant-derived terpenes as α -glucosidase inhibitors for the discovery of therapeutic agents against type 2 diabetes mellitus. South Afr J Bot 2021;143:462–73. <https://doi.org/10.1016/j.sajb.2021.09.030>
- [27] Feher M, Williams CI. Numerical errors in minimization based binding energy calculations. 3200–12 J Chem Inf Model 2012;52. <https://doi.org/10.1021/ci300298d>
- [28] Ponder JW, Case DA. Force Fields for Protein Simulations, 2003, p. 27–85. [https://doi.org/10.1016/S0065-3233\(03\)66002-X](https://doi.org/10.1016/S0065-3233(03)66002-X)
- [29] Wang J, Wolf RM, Caldwell JW, Kollman PA, Case DA. Development and testing of a general amber force field. J Comput Chem 2004;25:1157–74. <https://doi.org/10.1002/jcc.20035>
- [30] Abbasi S, Raza S, Azam SS, Liedl KR, Fuchs JE. Interaction mechanisms of a melatonergic inhibitor in the melatonin synthesis pathway. J Mol Liq 2016;221:507–17. <https://doi.org/10.1016/j.molliq.2016.06.034>
- [31] Kollman PA, Massova I, Reyes C, Kuhn B, Huo S, Chong L, et al. Calculating structures and free energies of complex molecules: combining molecular mechanics and continuum models. Acc Chem Res 2000;33:889–97. <https://doi.org/10.1021/ar000033j>
- [32] Graf R, Anzali S, Buenger J, Pfluecker F, Driller H. The multifunctional role of ectoine as a natural cell protectant. Clin Dermatol 2008;26:326–33. <https://doi.org/10.1016/j.clindermatol.2008.01.002>
- [33] Choi O, Cho J, Kang B, Lee Y, Kim J. Negatively regulated aerobactin and desferrioxamine E by fur in *Pantoea ananatis* are required for full siderophore production and antibacterial activity, but not for virulence. Appl Environ Microbiol 2022;88. <https://doi.org/10.1128/aem.02405-21>
- [34] Hung L-W, Wang IX, Nikaido K, Liu P-Q, Ames GF-L, Kim S-H. Crystal structure of the ATP-binding subunit of an ABC transporter. Nature 1998;396:703–7. <https://doi.org/10.1038/25393>
- [35] Staudenmaier H, Van Hove B, Yaraghi Z, Braun V. Nucleotide sequences of the fecBCDE genes and locations of the proteins suggest a periplasmic-binding-protein-dependent transport mechanism for iron(III) dicitrate in *Escherichia coli*. J Bacteriol 1989;171:2626–33. <https://doi.org/10.1128/jb.171.5.2626-2633.1989>
- [36] Hillerich B, Westpheling J. A new GntR family transcriptional regulator in *Streptomyces coelicolor* is required for morphogenesis and antibiotic production and controls transcription of an abc transporter in response to carbon source. J Bacteriol 2006;188:7477–87. <https://doi.org/10.1128/JB.00898-06>
- [37] Rigali S, Nothaft H, Noens EEE, Schlicht M, Colson S, Müller M, et al. The sugar phosphotransferase system of *Streptomyces coelicolor* is regulated by the GntR-family regulator DasR and links N-acetylglucosamine metabolism to the control of development. Mol Microbiol 2006;61:1237–51. <https://doi.org/10.1111/j.1365-2958.2006.05319.x>
- [38] Rigali S. Extending the classification of bacterial transcription factors beyond the helix-turn-helix motif as an alternative approach to discover new cis/trans relationships. Nucleic Acids Res 2004;32:3418–26. <https://doi.org/10.1093/nar/gkh673>
- [39] Matilla MA, Velando F, Martín-Mora D, Monteagudo-Cascales E, Krell T. A catalogue of signal molecules that interact with sensor kinases, chemoreceptors and transcriptional regulators. FEMS Microbiol Rev 2022;46. <https://doi.org/10.1093/femsre/fuab043>
- [40] van Aalten DMF, DiRusso CC, Knudsen J, Wierenga RK. Crystal structure of FadR, a fatty acid-responsive transcription factor with a novel acyl coenzyme A-binding fold. EMBO J 2000;19:5167–77. <https://doi.org/10.1093/emboj/19.19.5167>
- [41] Gabrio T, Böhme MWJ, Hack U, Miljanic T, Weidner U. Ambrosia – eine Ursache für die Zunahme von Allergien? Allergologie 2009;32:453–61. <https://doi.org/10.5414/ALP32453>
- [42] Ghose AK, Viswanadhan VN, Wendoloski JJ. A knowledge-based approach in designing combinatorial or medicinal chemistry libraries for drug discovery. 1. a qualitative and quantitative characterization of known drug databases. J Comb Chem 1999;1:55–68. <https://doi.org/10.1021/cc9800071>
- [43] Sanchez-Covarrubias L, Slosky LM, Thompson BJ, Zhang Y, Laracuent M-L, DeMarco KM, et al. P-glycoprotein modulates morphine uptake into the CNS: a role for the non-steroidal anti-inflammatory drug diclofenac. PLoS One 2014;9:e88516. <https://doi.org/10.1371/journal.pone.0088516>
- [44] Husain A, Makadia V, Valicherla GR, Riyazuddin M, Gayen JR. Approaches to minimize the effects of P-glycoprotein in drug transport: a review. Drug Dev Res 2022;83:825–41. <https://doi.org/10.1002/ddr.21918>
- [45] Wang X, Chen Y, Dahmani FZ, Yin L, Zhou J, Yao J. Amphiphilic carboxymethyl chitosan-quercetin conjugate with P-gp inhibitory properties for oral delivery of paclitaxel. Biomaterials 2014;35:7654–65. <https://doi.org/10.1016/j.biomaterials.2014.05.053>
- [46] Manzari MT, Shamay Y, Kiguchi H, Rosen N, Scaltriti M, Heller DA. Targeted drug delivery strategies for precision medicines. Nat Rev Mater 2021;6:351–70. <https://doi.org/10.1038/s41578-020-00269-6>
- [47] Gudín J. Opioid therapies and cytochrome P450 interactions. J Pain Symptom Manag 2012;44:S4–14. <https://doi.org/10.1016/j.jpainsymman.2012.08.013>
- [48] Mahizan NA, Yang S-K, Moo C-L, Song AA-L, Chong C-M, Chong C-W, et al. Terpene derivatives as a potential agent against antimicrobial resistance (AMR) Pathogens. Molecules 2019;24:2631. <https://doi.org/10.3390/molecules24142631>
- [49] Lorenzi V, Muselli A, Bernardini AF, Berti L, Pagès J-M, Amaral L, et al. Geraniol restores antibiotic activities against multidrug-resistant isolates from gram-negative species. Antimicrob Agents Chemother 2009;53:2209–11. <https://doi.org/10.1128/AAC.00919-08>
- [50] Jiménez-Arellanes A, Luna-Herrera J, Cornejo-Garrido J, López-García S, Castro-Mussot ME, Meckes-Fischer M, et al. Ursolic and oleanolic acids as antimicrobial and immunomodulatory compounds for tuberculosis treatment. BMC Complement Alter Med 2013;13:258. <https://doi.org/10.1186/1472-6882-13-258>
- [51] Bin Sayeed M, Karim S, Sharmin T, Morshed M. Critical analysis on characterization, systemic effect, and therapeutic potential of beta-sitosterol: a plant-derived orphan phytosterol. Medicines 2016;3:29. <https://doi.org/10.3390/medicines3040029>
- [52] Squeglia F, Romano M, Ruggiero A, Vitagliano L, De Simone A, Berisio R. Carbohydrate recognition by RpfB from *Mycobacterium tuberculosis* unveiled by crystallographic and molecular dynamics analyses. Biophys J 2013;104:2530–9. <https://doi.org/10.1016/j.bpj.2013.04.040>
- [53] Squeglia F, Romano M, Ruggiero A, Vitagliano L, De Simone A, Berisio R. Carbohydrate recognition by RpfB from *Mycobacterium tuberculosis* unveiled by crystallographic and molecular dynamics analyses. Biophys J 2013;104:2530–9. <https://doi.org/10.1016/j.bpj.2013.04.040>
- [54] Kerantzas CA, Jacobs WR. Origins of combination therapy for tuberculosis: lessons for future antimicrobial development and application. MBio 2017;8. <https://doi.org/10.1128/mBio.01586-16>
- [55] Compagne N, Vieira Da Cruz A, Müller RT, Hartkoorn RC, Flipo M, Pos KM. Update on the discovery of efflux pump inhibitors against critical priority gram-negative bacteria. Antibiotics 2023;12:180. <https://doi.org/10.3390/antibiotics12010180>